



4.2.2 The Method of Normal Equations ¶

04.2.2 Normal equations

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Consider again the LLS problem: Given $A \in \mathbb{C}^{m \times n}$ and $b \in \mathbb{C}^m$ find $\hat{x} \in \mathbb{C}^n$ such that

$$\|b - A\hat{x}\|_2 = \min_{x \in \mathbb{C}^n} \|b - Ax\|_2.$$

We list a sequence of observations that you should have been exposed to in previous study of linear algebra:

- $\hat{b} = A\hat{x}$ is in the column space of A .
- $\hat{b}$ equals the member of the column space of A that is closest to b , making it the orthogonal projection of b onto the column space of A .
- Hence the residual, $b - \hat{b}$, is orthogonal to the column space of A .
- From [Figure 4.2.1.2](#) we deduce that $b - \hat{b} = b - A\hat{x}$ is in $\mathcal{N}(A^H)$, the left null space of A .
- Hence $A^H(b - A\hat{x}) = 0$ or, equivalently,

$$A^H A \hat{x} = A^H b.$$

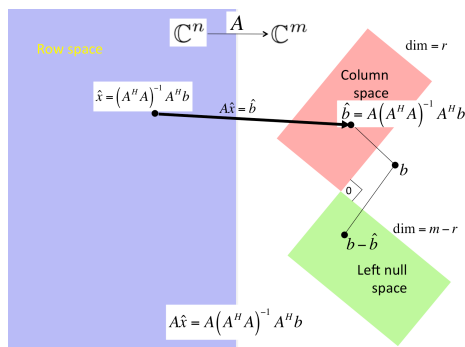
This linear system of equations is known as the normal equations.

- If A has linearly independent columns, then $\text{rank}(A) = n$, $\mathcal{N}(A) = \emptyset$, and $A^H A$ is nonsingular. In this case,

$$\hat{x} = (A^H A)^{-1} A^H b.$$

Obviously, this solution is in the row space, since $\mathcal{R}(A) = \mathbb{C}^n$.

With this, we have discovered what is known as the Method of Normal Equations. These steps are summarized in [Figure 4.2.2.1](#)



[images/Chapter04/FundamentalSpacesLLSLinIndep.pptx](#)

Figure 4.2.2.1. Solving LLS via the Method of Normal Equations when A has linearly independent columns (and hence the row space of A equals $\mathbb{C}^n$).

Definition 4.2.2.2. (Left) pseudo inverse. Let $A \in \mathbb{C}^{m \times n}$ have linearly independent columns. Then

$$A^\dagger = (A^H A)^{-1} A^H$$

is its (left) pseudo inverse.

Homework 4.2.2.1. Let $A \in \mathbb{C}^{m \times m}$ be nonsingular. Then $A^{-1} = A^\dagger$.

▼ [Solution](#)

$$AA^\dagger = A(A^H A)^{-1} A^H = AA^{-1} A^{-H} A^H = II = I.$$

Homework 4.2.2.2. Let $A \in \mathbb{C}^{m \times n}$ have linearly independent columns. ALWAYS/SOMETIMES/NEVER: $AA^\dagger = I$.

▼ [Hint](#) ▼ [Answer](#) ▼ [Solution](#)

An example where $AA^\dagger = I$ is the case where $m = n$ and hence A is nonsingular.

An example where $AA^\dagger \neq I$ is $A = e_0$ for $m > 1$. Then

$$\begin{aligned} &AA^\dagger \\ &= \text{< instantiate >} \\ &\begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix} \underbrace{\begin{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix}^H & \begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix} \end{pmatrix}^{-1}}_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix}^H \\ &= \text{< simplify >} \\ &\begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix} (1 \quad 0 \quad \dots) \\ &= \text{< multiply out >} \\ &\begin{pmatrix} 1 & 0 & \dots \\ 0 & 0 & \dots \\ \vdots & \vdots & \end{pmatrix} \\ &= \text{< } m > 1 \text{ >} \\ &\neq I. \end{aligned}$$

SOMETIMES

Consider $A = (e_0)$.

Ponder This 4.2.2.3. The last exercise suggests there is also a right pseudo inverse. How would you define it?